

Artificial intelligence in medical emergencies: what clinical trials are starting to show

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Abstract

Artificial intelligence (AI) is rapidly entering emergency medicine, promising faster triage, more accurate diagnosis, and better allocation of scarce resources. Yet its real benefit depends on clinical trials rather than technical performance alone. This article synthesizes prospective and randomized studies summarized in Cochrane-indexed and Cochrane-searching systematic reviews, alongside individual emergency department (ED) and prehospital trials, to examine how AI performs in real emergency settings. We focus on AI-driven triage, clinical decision support, and prehospital risk prediction, highlighting effects on response times, diagnostic accuracy, and patient outcomes. Across trials, AI tools frequently outperform traditional triage scores and standard decision support on accuracy-related metrics, but gains in hard clinical outcomes are modest and context-dependent. Methodological heterogeneity, limited randomization, and workflow integration issues temper current enthusiasm. Overall, AI appears most promising as an adjunct that accelerates and standardizes decision-making rather than as an autonomous system, underscoring the need for larger, pragmatic trials.

Keywords: artificial intelligence, emergency department, triage, decision support, prehospital care, machine learning, response time, diagnostic accuracy

Introduction

Emergency departments and ambulance systems around the world are under increasing pressure from rising patient volumes, crowding, and workforce shortages, creating strong incentives to use AI to prioritize patients and streamline decisions. Systematic searches that include the Cochrane Library show a rapidly expanding literature on AI and machine learning for ED triage, outcome prediction, and prehospital risk assessment, although relatively few studies are high-quality randomized trials. In ED triage, machine-learning models trained on electronic health record (EHR) data, vital signs, and free-text chief complaints have repeatedly outperformed conventional triage scales in predicting intensive care admission, mortality, or need for urgent intervention. Similarly, prehospital studies report that gradient boosting and related models can improve discrimination over early warning scores, potentially reducing under- and over-triage for trauma, stroke, and cardiac emergencies. Clinical decision support systems (CDSS) integrated into ED EHRs also show mostly positive effects on process measures and some patient-centered outcomes, although the evidence base is

dominated by pre–post designs. This article draws on these trials and the Cochrane-oriented review landscape to explain how AI actually performs in emergency care and what that means for clinicians and patients.

Methods

We narratively review clinical studies and systematic reviews that evaluated AI-based interventions in emergency medicine and were identified through reviews that explicitly searched the Cochrane Library or are indexed in databases commonly screened by Cochrane reviews (PubMed, Embase, CENTRAL). Priority was given to prospective and randomized trials conducted in real-world clinical environments, including ED triage systems, EHR-integrated CDSS, and prehospital decision tools. We used reported outcomes such as triage accuracy, area under the receiver operating characteristic curve (AUROC), time to decision, and patient-centered endpoints (e.g. mortality, critical care use) to compare interventions. Because formal pairwise meta-analysis was not consistently available for AI interventions, we focused on descriptive comparisons of study design, intervention type, and measured effects as presented in the trials and their summarizing reviews.

Results

Overview of AI interventions in emergency care

Systematic reviews of AI in ED triage and prehospital care describe a diverse set of tools: supervised machine-learning models for acuity scoring, deep-learning models for imaging or multimodal data, and rule-based or probabilistic CDSS for diagnosis and treatment support. Most ED triage tools use structured EHR variables such as age, vital signs, labs, and arrival mode, with several recent models adding free-text chief complaints through natural language processing. Prehospital systems rely heavily on vital signs, electrocardiogram features, and dispatch information, with gradient boosting and random forests being particularly common algorithms. Across these studies, AI is almost always deployed as a decision support layer: triage nurses and physicians remain responsible for final decisions but receive risk scores, alerts, or prioritized lists to guide action.

Comparative overview of key clinical trials

The table below summarizes representative prospective or trial-like studies drawn from Cochrane-searching reviews and individual reports, focusing on different emergency settings.

Selected AI interventions in emergency medicine

Setting & study type	Intervention	Comparator	Main outcomes reported
ED triage prospective implementations summarized by Jung et al. 2024,	– Machine-learning triage tools predicting high-acuity outcomes using EHR and vital signs	Standard nurse triage scales (e.g. 5-level acuity)	Triage accuracy 80.5–99.1%; reduced under-triage; variable effects on time to triage and downstream outcomes

ED decision prospective evaluation of XGBoost-based model	admission –	AI model updating admission predictions every 10 minutes, integrated in workflow	Usual physician decision-making without AI time aid	Reduction in time to admission decision by median 111 minutes; AUROC 0.89; no signal of increased safety events
ED CDSS systematic review of EHR-integrated tools,	–	Rule-based and probabilistic CDSS for imaging, medication, and disease-specific care	Usual care or pre-implementation periods	83% of 42 studies showed improved process or clinical outcomes; only 3 randomized trials, mostly small and disease-specific
Pediatric CDSS – controlled study	CHD	Condition-specific decision support for children with congenital heart disease in ED	Standard ED care without CHD-specific CDSS	Improved sense-making and management decisions; more appropriate tests and interventions; limited change in correct final diagnosis rate
Prehospital risk prediction – ML vs NEWS-type scores,,		Gradient boosting and other ML models predicting short-term adverse outcomes	National Warning (NEWS/NEWS2) and local triage tools	Higher AUROC and precision-recall performance; reduced false positives at matched sensitivity; simulated reductions in over-triage and resource misallocation

Effects on response times and operational metrics

AI's most consistent real-world benefit in emergency care lies in timing and flow rather than dramatic improvements in survival. In a prospective ED study, an admission-prediction model using extreme gradient boosting updated risk scores every 10 minutes and shortened time to admission decisions by a median of 111 minutes for correctly predicted patients, without evidence of safety compromise. Reviews of machine-learning in emergency medical services (EMS) describe operational tools that optimize ambulance deployment, with some algorithms cutting expected response times by around 1.5 minutes compared with manually designed strategies, although these findings come largely from simulation or retrospective evaluations rather than randomized trials. Within the ED, real-time AI triage can reduce bottlenecks by flagging high-risk patients earlier, but studies summarized in recent prospective triage reviews report mixed effects on overall ED length of stay because downstream constraints such as bed availability remain limiting. As a result, AI appears more reliable today as a way to speed decision-making than as a guaranteed means to shorten total episode-of-care duration.

Diagnostic accuracy and triage performance

Across ED and prehospital studies, AI tools frequently exceed traditional scores in predicting critical outcomes, but gains vary by setting and data richness. A 2024 systematic review of AI for ED triage, which searched the Cochrane Library among other databases, found that machine-learning models generally achieved higher accuracy, sensitivity, and specificity than conventional triage scales for predicting hospital admission, intensive care need, or mortality. A prospective synthesis of implemented AI triage systems reported prediction accuracies between 80.5% and 99.1%, with reductions in under-triage and better discrimination of high-risk patients when free-text data were included. In prehospital settings, gradient boosting and Bayesian networks trained on ambulance records outperformed NEWS2 and national triage scores in AUROC and precision-recall analyses, suggesting that nonlinear models capture risk patterns missed by linear scores. However, an ED review of EHR-integrated CDSS showed that while 83% of studies reported positive effects, many gains were in process measures like more appropriate imaging or medication ordering rather than direct improvements in diagnostic accuracy or mortality.

Patient outcomes and safety

Evidence that AI meaningfully improves hard patient outcomes in emergency medicine remains limited but encouraging. Scoping and systematic reviews of prehospital AI suggest that improved risk prediction and destination planning could translate into fewer secondary transfers and faster access to definitive care for trauma, stroke, and cardiac emergencies, but most outcome data are modeled rather than trial-based. In ED settings, disease-specific CDSS, such as tools for managing children with congenital heart disease, have improved decision quality—more appropriate tests and interventions—without clear, immediate changes in mortality, partly due to small sample sizes and short follow-up. The admission-prediction trial mentioned earlier demonstrated that accelerating disposition decisions did not increase adverse events, implying at least non-inferior safety while improving throughput. Overall, current trials support AI as a safe adjunct when carefully implemented, but definitive evidence that it reduces deaths or major complications at scale is still emerging, and Cochrane-style high-quality randomized trials are relatively scarce in this domain.

Discussion

Systematic reviews that include the Cochrane Library paint a picture of rapid growth in AI applications for emergency medicine, but also of evidence that is fragmented across settings and outcome types. ED triage trials and prospective implementations consistently show that machine-learning models can classify risk more accurately than traditional scales, especially when they combine structured clinical data with natural language features from patient complaints. Prehospital studies similarly indicate that AI can refine triage and destination decisions, outperforming early warning scores and local rules in predicting short-term deterioration and critical care needs. Yet, despite these performance gains, the translation into improved mortality or major morbidity remains incompletely demonstrated, emphasizing the need for pragmatic, multi-center

randomized trials registered in platforms like the Cochrane Central Register of Controlled Trials.

Another theme across Cochrane-searching reviews is that study design quality lags behind the speed of technological innovation. Many ED CDSS evaluations use pre-post or quasi-experimental designs, with only a handful of randomized trials, and even fewer focused specifically on AI-driven algorithms rather than rule-based alerts. Prospective triage implementations often lack concurrent randomized controls, making it difficult to separate AI effects from secular changes in staffing or crowding. Prehospital studies tend to be retrospective, using historical ambulance data, and validate models in silico without testing them in live dispatch or field workflows. Cochrane-style standards for bias assessment and outcome reporting are only beginning to be applied rigorously to AI trials, highlighting an important methodological gap.

Despite these limitations, the convergence of results across emergency settings suggests several practical implications for the future of emergency medicine. First, AI is most promising as an augmentation tool that enhances clinicians' situational awareness and speeds up complex decisions, rather than as a stand-alone "black box" that replaces professional judgment. Second, workflow integration—such as continuously updated risk scores embedded in ED dashboards or ambulance tablets—is critical; models that perform well on historical data may fail if they disrupt established team routines or overload clinicians with non-actionable alerts. Third, equity and generalizability need greater attention: many studies are single-center and high-income country-based, and biased training data could worsen triage for under-represented groups if not carefully addressed. Finally, future Cochrane reviews will likely demand not only better study designs but also transparent reporting of model development, calibration, and updating, so that clinicians and policymakers can weigh AI tools alongside other emergency interventions using familiar evidence-based frameworks.

Conclusion

AI is beginning to change emergency medicine not through spectacular autonomous systems, but through quieter, embedded tools that help clinicians triage faster, decide sooner, and focus attention where it matters most. Current clinical trials and Cochrane-oriented systematic reviews show that AI-based triage and decision support often outperform traditional scores and rules in predicting who is sickest, while maintaining at least comparable safety. However, improvements in response times and diagnostic accuracy have not yet fully translated into robust, reproducible gains in survival or long-term outcomes, largely because existing studies are small, heterogeneous, and rarely randomized. To fulfill AI's promise in medical emergencies, the next wave of research must pair sophisticated algorithms with rigorous trial designs, transparent evaluation, and careful integration into the realities of crowded emergency departments and stretched ambulance services. For clinicians and patients

alike, the most exciting prospect is not AI replacing human expertise, but AI-enabled teams delivering faster, safer, and more equitable emergency care when minutes truly matter.

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